Impact of hospitals' LGBTQ+ inclusion efforts on patient satisfaction from 2016 to 2023: a retrospective longitudinal observational study

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► Additional supplemental material is published online only. To view, please visit the journal online (https://doi.org/10.1136/bmjqs-2024-018235).

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Received 4 November 2024
Accepted 9 February 2025

Check for updates

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To cite: Yu H, McHugh MD, Bauermeister JA, *et al*. *BMJ Qual Saf* Epub ahead of print: [please include Day Month Year]. doi:10.1136/bmjqs-2024-018235

ABSTRACT

Background Given the emphasis on promoting inclusive policies, we investigated the relationship between US hospitals' inclusion efforts for lesbian, gay, bisexual, transgender, queer or questioning, and other sexual and gender-diverse (LGBTQ+) populations and patient satisfaction from 2016 to 2023.

Methods This retrospective longitudinal observational study analysed 6 years of data between 2016 and 2023 from the Healthcare Equality Index (HEI), which measures hospitals' LGBTQ+ inclusion efforts, and the Hospital Consumer Assessment of Healthcare Providers and Systems, which measures patient satisfaction. Generalised estimating equations (GEE) were used to obtain population-averaged estimates of the association between hospitals' LGBTQ+ inclusion efforts—assessed by (1) their participation and (2) performance in the HEI (range: 0–100)—and patient satisfaction—measured by (1) patients' hospital rating (range: 0–100) and (2) willingness to recommend the hospital (range: 0–100). We accounted for hospital characteristics, including medical teaching status, specialised service capability, hospital size, ownership, system membership, region and metropolitan location.

Results Compared with hospitals that never participated in the HEI, those that occasionally participated reported a 0.33-point higher patient rating (p=0.019, 95% CI 0.05, 0.60) and a 0.49-point higher patient recommendation score (p=0.011, 95% CI 0.11, 0.87). Those who always participated reported a 1.30-point higher rating (p<0.001, 95% CI 0.89, 1.70) and a 1.90-point higher recommendation score (p<0.001, 95% CI 1.36, 2.44). Among hospitals that participated in the HEI, a 10-point increase in the total HEI score was associated with a 0.10-point increase in patient ratings (p=0.031, 95% CI 0.01, 0.20) and a 0.15-point increase in patient recommendations (p=0.023, 95% CI 0.02, 0.28).

Conclusion Hospitals engaging in LGBTQ+ inclusion efforts are associated with higher patient satisfaction.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Despite the emphasis on inclusive policies to promote equitable healthcare environments, limited research has specifically examined the impact of hospitals' inclusion efforts for lesbian, gay, bisexual, transgender, queer or questioning, and other sexual and gender-diverse (LGBTQ+) populations on patient satisfaction.

WHAT THIS STUDY ADDS

⇒ Consistent participation in and strong performance on the Healthcare Equality Index is associated with higher patient ratings and recommendations.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ The results can inform policy decisions that encourage LGBTQ+ inclusivity and underscore the need for ongoing research to assess the long-term impacts of organisational inclusion on diverse patient populations.

INTRODUCTION

Lesbian, gay, bisexual, transgender, queer or questioning and other sexual and gender diverse (LGBTQ+) individuals comprise an average of 9% of adults across 30 countries.[1] Despite social progress, they continue to face disproportionately high levels of discrimination in healthcare, which contribute to delayed care, poor health outcomes and mistrust in healthcare,[2–4] deepening disparities for LGBTQ+ individuals compared with non-LGBTQ+ counterparts.[5]

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In response to these ongoing disparities, the American Hospital Association (AHA) encourages the adoption of inclusive policies for patients and employees.[6] The Healthcare Equality Index (HEI), administered by the Human Rights Campaign Foundation, is a crucial tool for evaluating LGBTQ+ inclusion efforts in voluntarily participating hospitals, including aspects such as LGBTQ+ inclusive clinical services.[7] Hospitals often use HEI scores to set targets for improving inclusivity, benchmark performance against peer institutions and guide the implementation of specific policies and practices aimed at fostering an equitable environment for both patients and staff.[7] However, there is a lack of research examining its impact on outcomes, including patient satisfaction.

Patient satisfaction with hospital care in the USA is often assessed through two key global items from the Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS)[8]: overall hospital rating and patients' willingness to recommend the hospital.[9–11] Two studies found that hospitals with HEI Leader designations (awarded for highest performance) had higher HCAHPS patient ratings and patient recommendations compared with hospitals without such designations.[11 12] However, these studies were limited to two specific years and did not examine the impact of engaging in the HEI, which is voluntary, or the hospitals' performance across different HEI domains. It remains unclear whether any engagement in the HEI, regardless of final designation, impacts on patient experiences.

Theoretical framework

By fostering inclusion, organisations can improve culture, enhance employee well-being and elevate performance, including customer satisfaction, based on theories on inclusive organisations.[13 14] In healthcare settings, LGBTQ+ inclusion efforts could lead to higher HCAHPS scores through two key mechanisms. First, LGBTQ+ patients may recognise and value these inclusion efforts, leading to higher ratings on HCAHPS surveys. Second, an organisational commitment to LGBTQ+ inclusion may reflect a broader culture of equity and inclusion that benefits all patients, fostering improvements in care delivery and enhancing the overall patient experience.

Using longitudinal data, this study investigated the relationship between patient satisfaction and (1) hospitals' participation in the HEI (indicating LGBTQ+ inclusion efforts) and (2) hospitals' performance in the HEI (reflecting the extent of inclusion) over 6 years between 2016 and 2023.

Hypotheses

1. Hospitals' participation in the HEI is positively associated with patient satisfaction.
2. Among hospitals participating in the HEI, higher performance is positively associated with patient satisfaction.

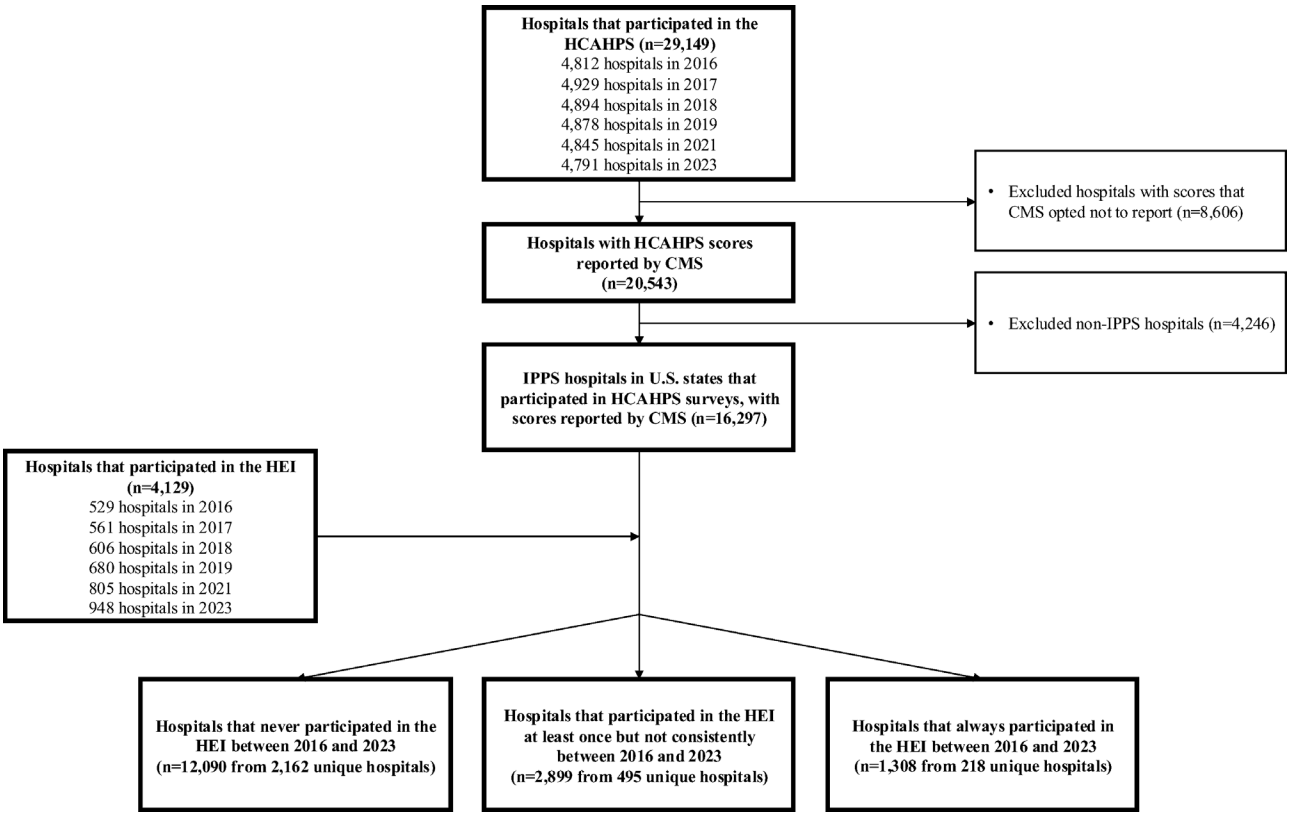

Figure 1 Sampling scheme. CMS, Centres for Medicare and Medicaid Services; HCAHPS, Hospital Consumer Assessment of Healthcare Providers and Systems; HEI, Healthcare Equality Index; IPPS, Inpatient Prospective Payment System.

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Table 1 Characteristics of hospitals based on participation in the Healthcare Equality Index from 2016 to 2023

	Overall (n=2875)	Never participated (n=2162)	Occasionally participated (n=495)	Always participated (n=218)	P value
Overall hospital rating (mean (SD))*	87.88 (3.5)	87.82 (3.7)	87.82 (2.9)	88.62 (3.0)	<0.001
Overall hospital rating (range)	55–98	55–98	69–97	75–96	
Patients' willingness to recommend the hospital (mean (SD))*	87.25 (4.7)	87.01 (4.9)	87.52 (3.9)	88.89 (3.9)	<0.001
Patients' willingness to recommend the hospital (range)	43–99	43–99	62–97	73–98	
Total Healthcare Equality Index score (mean (SD)*	90.51 (16.5)	Not available	83.65 (19.9)	96.74 (8.8)	<0.001
Non-discrimination and staff training subscore (mean (SD))†	38.56 (4.5)	Not available	37.40 (6.0)	39.61 (2.0)	<0.001
Patient services and support subscore (mean (SD))‡	26.17 (7.9)	Not available	23.31 (9.6)	28.76 (4.5)	<0.001
Employee benefits and policies subscore (mean (SD))§	17.78 (4.4)	Not available	16.35 (5.2)	19.08 (2.8)	<0.001
Patient and community engagement subscore (mean (SD))¶	8.13 (3.3)	Not available	6.84 (3.8)	9.31 (2.1)	<0.001
Medical teaching status (n (%))					<0.001
Major	256 (8.9)	93 (4.3)	78 (15.8)	85 (39.0)	
Minor	1068 (37.1)	845 (39.1)	163 (32.9)	60 (27.5)	
None	1551 (54.0)	1224 (56.6)	254 (51.3)	73 (33.5)	
Hospital size (n (%))					<0.001
Small (up to 100 beds)	970 (33.7)	849 (39.3)	99 (20.0)	22 (10.1)	
Medium (101–250 beds)	989 (34.4)	765 (35.4)	162 (32.7)	62 (28.4)	
Large (over 250 beds)	916 (31.9)	548 (25.3)	234 (47.3)	134 (61.5)	
Specialised service capability (n (%))					<0.001
Low capability	1944 (67.6)	1565 (72.4)	277 (56.0)	102 (46.8)	
Region (n (%))					<0.001
Midwest	671 (23.3)	537 (24.8)	95 (19.2)	39 (17.9)	
Northeast	446 (15.5)	230 (10.6)	154 (31.1)	62 (28.4)	
South	1171 (40.7)	1025 (47.4)	113 (22.8)	33 (15.1)	
West	587 (20.4)	370 (17.1)	133 (26.9)	84 (38.5)	
Metropolitan location (n (%))					<0.001
Metropolitan	2228 (77.5)	1565 (72.4)	448 (90.5)	215 (98.6)	
Ownership (n (%))					<0.001
Not-for-profit	2261 (78.6)	1580 (73.1)	464 (93.7)	217 (99.5)	
System membership (n (%))					<0.001
Yes	2218 (77.1)	1578 (73.0)	444 (89.7)	196 (89.9)	

*Score ranges from 0 to 100.
†Subscore ranges from 0 to 40.
‡Subscore ranges from 0 to 30.
§Subscore ranges from 0 to 20.
¶Subscore ranges from 0 to 10.

METHODS

This was a retrospective longitudinal observational study using US hospital data.

Data sources

We linked hospital data from three survey sources: the HEI, HCAHPS and AHA surveys. The HEI survey evaluates the LGBTQ+ inclusivity of policies and practices in healthcare facilities that voluntarily complete a detailed online survey and supporting documentation to validate responses.[7] The HCAHPS survey is required for hospitals that participate in the Medicare programme. It captures patient perspectives on hospital care quality through random sampling of discharged patients, which are then aggregated and publicly reported to compare hospital performance.[8] The AHA Annual Survey collects data on hospital characteristics and operations, including staffing and services.[15]

The HEI was conducted annually until 2020 when it transitioned to a biennial schedule. As a result, the HCAHPS data from 2020 and 2022 were excluded from analysis because HEI data were not available for those years, precluding evaluation of hospitals' participation or performance during those periods. This study focused on years where both HCAHPS and HEI data were collected—2016, 2017, 2018, 2019, 2021 and 2023.

Study sample

Given the repeated measures nature of the HEI and HCAHPS surveys, each *observation* reflects data from a total number of hospitals that overlap across the 6

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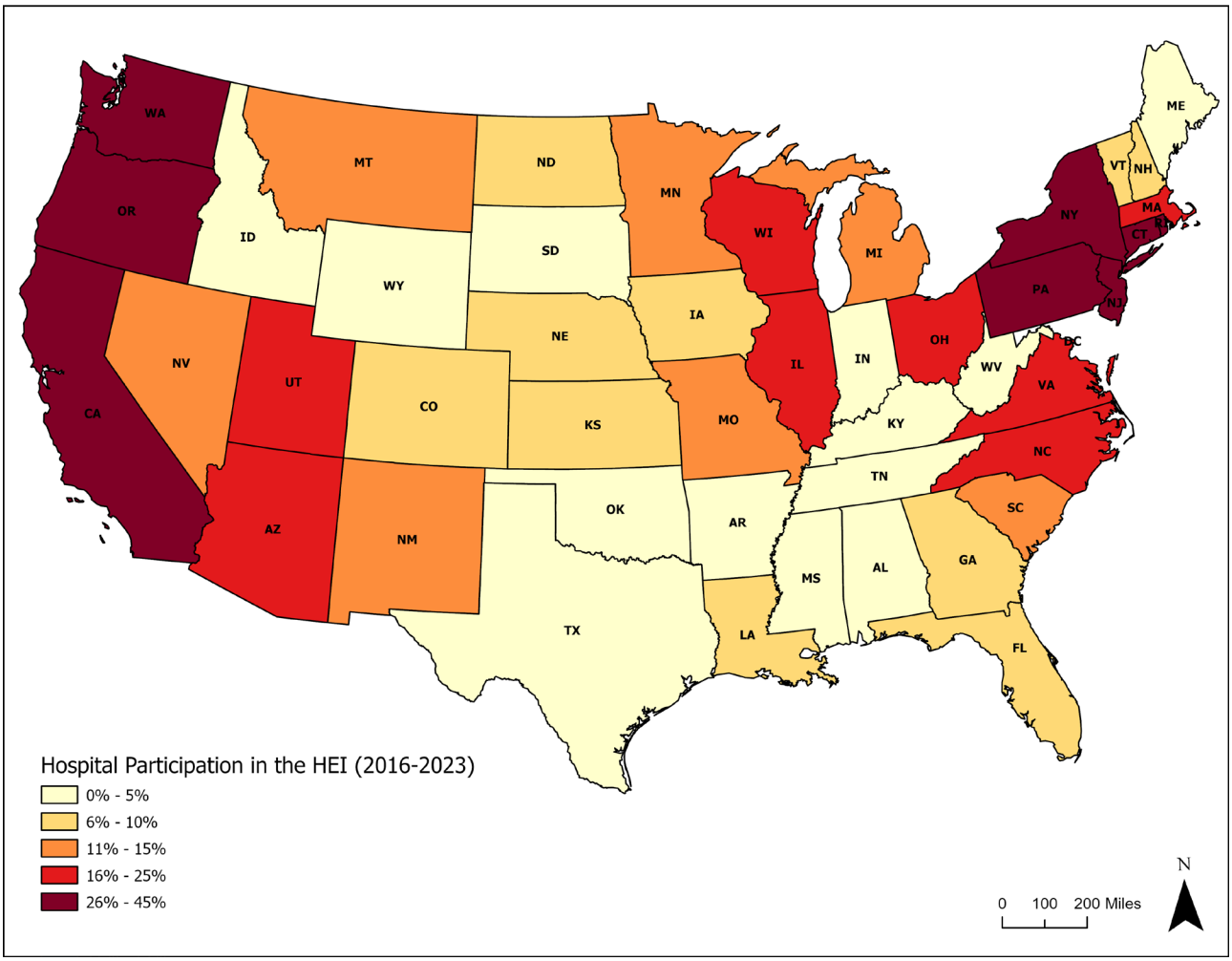

Figure 2 Hospitals that participated in the Healthcare Equality Index (HEI) in each state between 2016 and 2023. The percentage was calculated by dividing the number of hospitals that participated in the HEI in each state by the number of hospitals that participated in the Hospital Consumer Assessment of Healthcare Providers and Systems in that state for the years 2016, 2017, 2018, 2019, 2021 and 2023, then multiplying by 100. High-participation states were clustered in the Northeast and West regions.

years, while *unique hospitals* refer to distinct hospitals represented in those observations.

Inclusion criteria were: (1) participation in HCAHPS during any of the 6 years with linear mean scores publicly reported by the Centers for Medicare and Medicaid Services (CMS), (2) participation in the Inpatient Prospective Payment System (IPPS), a reimbursement system under Medicare that pays hospitals a predetermined amount per inpatient stay based on diagnosis-related groups and (3) participation in the AHA Annual Surveys in 2020, 2021 or 2022. CMS calculates and reports linear mean scores of HCAHPS measures to robustly compare hospitals with specific rules for the public reporting of these scores. Hospitals with fewer than 100 completed surveys over a four-quarter period or those not participating in the Inpatient Quality Reporting programmes (online supplemental appendix A) are exempt from reporting, thus excluded from our analysis. For hospitals that participated in the AHA surveys from 2020 to 2022 but had missing values, these values were retrieved from earlier AHA survey data.

Participation in the HEI and patient satisfaction (hypothesis 1)

To assess the relationship between HEI participation and HCAHPS scores, we applied our inclusion criteria to obtain the analytic sample (figure 1). The final sample for this hypothesis comprised three groups: hospitals that never participated in the HEI between 2016 and 2023 (*never participated*: 12 090 observations from 2162 unique hospitals), hospitals that participated in the HEI at least once but not consistently between 2016 and 2023 (*occasionally participated*: 2899 observations from 495 unique hospitals) and hospitals that always participated in the HEI between 2016 and 2023 (*always participated*: 1308 observations from 218 unique hospitals).

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Table 2 Multivariable analysis examining the association between hospitals' participation in the Healthcare Equality Index and two patient satisfaction outcomes from 2016 to 2023 (n=2875 unique hospitals with 16 297 observations over the 6-year period)

Overall rating of hospital			
Predictor variable and covariates	**Unstandardised coefficients (95% CI)**	**Standardised coefficients (95% CI)**	**P value**
Participation in the Healthcare Equality Index			<0.001
Never participated (reference)			
Occasionally participated	0.33 (0.05, 0.60)	0.09 (0.02, 0.17)	0.019
Always participated	1.30 (0.89, 1.70)	0.37 (0.25, 0.49)	<0.001
Region			<0.001
South (reference)			
Midwest	0.37 (0.10, 0.63)	0.10 (0.03, 0.18)	0.007
Northeast	−0.97 (−1.32, −0.61)	−0.28 (−0.38, −0.18)	0.001
West	−0.34 (−0.65, −0.04)	−0.10 (−0.18, −0.01)	0.030
Metropolitan location			
Metropolitan (reference)			
Non-metropolitan	−0.80 (−1.09, −0.52)	−0.23 (−0.31, −0.15)	<0.001
Medical teaching status			<0.001
None (reference)			
Minor	−1.09 (−1.33, −0.85)	−0.31 (−0.38, −0.24)	<0.001
Major	−0.90 (−1.30, −0.49)	−0.26 (−0.37, −0.14)	<0.001
Hospital size			<0.001
Small (up to 100 beds) (reference)			
Medium (101–250 beds)	−2.02 (−2.31, −1.72)	−0.58 (−0.66, −0.49)	<0.001
Large (over 250 beds)	−2.37 (−2.73, −2.01)	−0.68 (−0.78, −0.57)	<0.001
Specialised service capability			
High capability	1.03 (0.76, 1.30)	0.29 (0.22, 0.37)	<0.001
Low capability (reference)			
Ownership			
Not-for-profit (reference)			
For-profit	−0.89 (−1.21, −0.57)	−0.25 (−0.34, −0.16)	<0.001
System membership			
Yes (reference)			
No	0.50 (0.20, 0.81)	0.14 (0.06, 0.23)	0.001
Year of data collection			<0.001
2016 (reference)			
2017	−0.33 (−0.39, −0.27)	−0.09 (−0.11, −0.08)	<0.001
2018	−0.63 (−0.70, −0.56)	−0.18 (−0.20, −0.16)	<0.001
2019	−0.53 (−0.60, −0.45)	−0.15 (−0.17, −0.13)	<0.001
2021	−1.68 (−1.77, −1.58)	−0.48 (−0.51, −0.45)	<0.001
2023	−1.87 (−1.98, −1.76)	−0.53 (−0.56, −0.50)	<0.001
Willingness to recommend the hospital			
Predictor variable and covariates	**Unstandardised coefficients (95% CI)**	**Standardised coefficients (95% CI)**	**P value**
Participation in the Healthcare Equality Index			<0.001
Never participated (reference)			
Occasionally participated	0.49 (0.11, 0.87)	0.11 (0.02, 0.19)	0.011
Always participated	1.90 (1.36, 2.44)	0.41 (0.29, 0.52)	<0.001
Region			<0.001
South (reference)			
Midwest	0.42 (0.05, 0.79)	0.09 (0.01, 0.17)	0.026
Northeast	−0.90 (−1.38, −0.42)	−0.19 (−0.29, −0.09)	<0.001
West	0.22 (−0.65, 0.20)	−0.05 (−0.14, 0.04)	0.304
Metropolitan location			
Metropolitan (reference)			
Non-metropolitan	−1.82 (−2.21, −1.43)	−0.39 (−0.47, −0.31)	<0.001
Medical teaching status			<0.001
None (reference)			
Minor	−1.42 (−1.75, −1.09)	−0.30 (−0.38, −0.23)	<0.001
Major	−1.16 (−1.70, −0.61)	−0.25 (−0.36, −0.13)	<0.001
Hospital size			<0.001
Small (up to 100 beds) (reference)			
Medium (101–250 beds)	−2.33 (−2.74, −1.92)	−0.50 (−0.59, −0.41)	<0.001

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Table 2 Continued

Willingness to recommend the hospital			
Predictor variable and covariates	**Unstandardised coefficients (95% CI)**	**Standardised coefficients (95% CI)**	**P value**
Large (over 250 beds)	−2.53 (−3.02, −2.03)	−0.54 (−0.65, −0.43)	<0.001
Specialised service capability			
High capability	1.73 (1.36, 2.09)	0.37 (0.29, 0.45)	<0.001
Low capability (reference)			
Ownership			
Not-for-profit (reference)			
For-profit	−1.31 (−1.75, −0.88)	−0.28 (−0.37, −0.19)	<0.001
System membership			
Yes (reference)			
No	0.54 (0.13, 0.96)	0.12 (0.03, 0.20)	0.010
Year of data collection			<0.001
2016 (reference)			
2017	−0.04 (−0.11, 0.03)	−0.01 (−0.02, 0.01)	0.245
2018	−0.28 (−0.36, −0.20)	−0.06 (−0.08, −0.04)	<0.001
2019	−0.32 (−0.41, −0.23)	−0.07 (−0.09; −0.05)	<0.001
2021	−1.74 (−1.86, −1.62)	−0.37 (−0.40, −0.35)	<0.001
2023	−2.20 (−2.34, −2.06)	−0.47 (−0.50, −0.44)	<0.001

Degree of performance in the HEI and patient satisfaction (hypothesis 2)

To assess the relationship between hospitals' performance in the HEI and HCAHPS scores, we examined hospitals that participated in the HEI for at least 1 year from the final sample for the previous measure, comprising 2496 observations from 713 distinct hospitals.

Measures

Predictor variable I—participation in the HEI (hypothesis 1)

To capture the consistency and commitment of hospitals' LGBTQ+ inclusion efforts, we used HEI participation as a predictor for each year. This variable was binary for each year, with a value of 0 for hospitals that did not participate that year and 1 for those that did.

Predictor variable II—performance in the HEI (hypothesis 2)

To capture the extent of inclusion, we used hospitals' annual HEI total scores (0–100) as a predictor, focusing on hospitals that participated in the HEI. This total score consists of four subscores. *Non-discrimination and staff training* evaluates non-discrimination policies for patients and employees and staff training on LGBTQ+ inclusive care (up to 40 points). *Patient services and support* assesses LGBTQ+-specific clinical services, including gender-affirming care (up to 30 points). *Employee benefits and policies* covers support for LGBTQ+ employees, such as health benefits (up to 20 points). *Patient and community engagement* measures outreach to the LGBTQ+ community through events (up to 10 points). Points may be deducted for discriminatory practices or anti-LGBTQ+ incidents.[16] In our primary analysis, we used total HEI scores. For sensitivity analyses, we examined each of the four subscores to determine specific associations with patient satisfaction.

Outcome variable

Patient satisfaction with hospital care included two measures, each converted to a score out of 100: overall hospital rating and patients' willingness to recommend the hospital. Higher scores indicate better ratings and greater willingness to recommend, both based on a weighted average of responses (online supplemental appendix B).

Covariates

We included covariates that may impact the predictor and outcome variables but were likely unchanged (eg, time-invariant) throughout the study period. These included medical teaching status, specialised service capability, hospital size, census region, metropolitan location, hospital ownership and system membership (online supplemental appendix C).

Statistical analysis

We conducted a descriptive analysis based on hospitals' HEI participation from 2016 to 2023, categorised as *never*, *occasionally* or *always* participated. HCAHPS and HEI scores were analysed with analysis of variance, and categorical hospital characteristics with the χ^2 test. To account for within-subject correlation in repeated measures and obtain population-averaged (or marginal) estimates, we applied a generalised estimating equation (GEE) model.[17 18] The GEE was used to make inferences about the average effect of the predictors on the outcome across the entire population, rather than focusing on the effect within individual hospitals.[19]

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Table 3 Multivariable analysis examining the association between hospitals' performance in the Healthcare Equality Index and two patient satisfaction outcomes from 2016 to 2023 (n=713 unique hospitals with 2496 observations over the 6-year period)

Overall rating of hospital			
Predictor variable and covariates	**Unstandardised coefficients (95% CI)**	**Standardised coefficients (95% CI)**	**P value**
Total Healthcare Equality Index score	0.10 (0.01, 0.20)	0.06 (0.01, 0.11)	0.031
Region			<0.001
South (reference)			
Midwest	0.26 (−0.36, 0.87)	0.09 (-0.12, 0.29)	0.412
Northeast	−1.35 (−1.97, −0.73)	−0.45 (-0.66, −0.25)	<0.001
West	0.31 (−0.29, 0.92)	0.10 (-0.10, 0.31)	0.315
Metropolitan location			
Metropolitan (reference)			
Non-metropolitan	−0.48 (−1.54, 0.58)	−0.16 (-0.52, 0.19)	0.373
Medical teaching status			0.369
None (reference)			
Minor	−0.20 (−0.70, 0.29)	−0.16 (-0.38, 0.06)	0.422
Major	−0.47 (−1.13, 0.19)	−0.07 (-0.23, 0.10)	0.161
Hospital size			0.006
Small (up to 100 beds) (reference)			
Medium (101–250 beds)	−0.62 (−1.29, 0.06)	−0.21 (-0.43, 0.02)	0.073
Large (over 250 beds)	−1.26 (−2.05, −0.48)	−0.42 (-0.69, −0.16)	0.002
Specialised service capability			
High capability	1.22 (0.63, 1.80)	0.41 (0.21, 0.61)	<0.001
Low capability (reference)			
Ownership			
Not-for-profit (reference)			
For-profit	−2.30 (−3.22, −1.37)	−0.77 (-1.08, −0.46)	<0.001
System membership			
Yes (reference)			
No	−0.65 (−1.48, 0.18)	−0.22 (-0.50, 0.06)	0.127
Year of data collection			<0.001
2016 (reference)			
2017	0.10 (−0.09, 0.28)	0.03 (-0.03, 0.09)	0.311
2018	−0.20 (−0.43, 0.04)	−0.07 (-0.15, 0.01)	0.106
2019	−0.05 (−0.33, 0.23)	−0.02 (-0.11, 0.08)	0.728
2021	−1.06 (−1.37, −0.75)	−0.36 (-0.46, −0.25)	<0.001
2023	−1.63 (−1.96, −1.31)	−0.55 (-0.66, −0.44)	<0.001
Willingness to recommend the hospital			
Predictor variable and covariates	**Unstandardised coefficients (95% CI)**	**Standardised coefficients (95% CI)**	**P value**
Total Healthcare Equality Index score	0.15 (0.02, 0.28)	0.06 (0.01, 0.12)	0.023
Region			<0.001
South (reference)			
Midwest	0.08 (-0.79, 0.94)	0.02 (−0.20, 0.24)	0.862
Northeast	−1.58 (-2.42, −0.73)	−0.40 (−0.61, −0.19)	<0.001
West	0.44 (-0.38, 1.25)	0.11 (−0.10, 0.32)	0.293
Metropolitan location			
Metropolitan (reference)			
Non-metropolitan	−1.66 (-3.09, −0.24)	−0.42 (−0.78, −0.06)	0.022
Medical teaching status			0.484
None (reference)			
Minor	−0.29 (-0.96, 0.38)	−0.07 (−0.24, 0.10)	0.394
Major	−0.51 (-1.38, 0.36)	−0.13 (−0.35, 0.09)	0.251
Hospital size			0.075
Small (up to 100 beds) (reference)			
Medium (101–250 beds)	−0.61 (-1.56, 0.33)	−0.16 (−0.40, 0.08)	0.205
Large (over 250 beds)	−1.23 (-2.31, −0.15)	−0.31 (−0.58, −0.04)	0.025
Specialised service capability			
High capability	1.80 (1.01, 2.59)	0.46 (0.26, 0.66)	<0.001
Low capability (reference)			
Ownership			
Not-for-profit (reference)			

Continued

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Table 3 Continued

Willingness to recommend the hospital			
Predictor variable and covariates	**Unstandardised coefficients (95% CI)**	**Standardised coefficients (95% CI)**	**P value**
For-profit	−3.06 (-4.27, −1.86)	−0.78 (−1.08, −0.47)	<0.001
System membership			
Yes (reference)			
No	−0.89 (−1.98, 0.21)	−0.23 (−0.50, 0.05)	0.112
Year of data collection			<0.001
2016 (reference)			
2017	0.12 (−0.12, 0.37)	0.03 (−0.03, 0.09)	0.319
2018	−0.24 (−0.55, 0.06)	−0.06 (−0.14, 0.02)	0.120
2019	−0.17 (−0.53, 0.19)	−0.04 (−0.13, 0.05)	0.354
2021	−1.38 (−1.77, −1.00)	−0.35 (−0.45, −0.25)	<0.001
2023	−2.27 (−2.69, −1.84)	−0.57 (−0.68, −0.47)	<0.001

The outcome variable in both GEE models was the HCAHPS score, with either HEI participation group or HEI score as the primary predictor. The covariates included hospital size, medical teaching status, specialised service capability, census region, metropolitan location, hospital ownership, system membership and year. The variance inflation factors used to check multicollinearity among the independent variables were low to moderate, ranging from 1.07 to 2.50. Our GEE model used a Gaussian family for continuous, symmetrically distributed outcomes, an identity link to model associations without transforming the outcome scale, and an independent correlation structure to account for correlations between repeated measures. We used robust SEs to address heteroscedasticity and other potential violations of regression assumptions.[20 21] Total HEI scores were used for the primary analysis, with HCAHPS as the outcome, and four HEI subscores were used for sensitivity analysis. We conducted our analyses using Stata (V.17, StataCorp, College Station, Texas, USA).

RESULTS

Among the 16 297 observations (2875 unique hospitals), 12 090 observations (2162 unique hospitals) were in the *never participated* group, 2899 observations (495 unique hospitals) were in the *occasionally participated* group and 1308 observations (218 unique hospitals) were in the *always participated* group (table 1). Hospitals that participated in the HEI between 2016 and 2023 were primarily clustered in the Northeast and West regions (figure 2). Differences were observed across all hospital-level covariates among the never, occasionally and always participated groups. Moreover, the mean total HEI scores differed significantly between the occasionally participated group (M=83.65, SD=19.9) and the always participated group (M=96.74, SD=8.8; t=−20.9, p<0.001).

Participation in the HEI (hypothesis 1)

Compared with hospitals that never participated in the HEI, those that occasionally participated reported a 0.33-point higher patient rating (p=0.019, 95% CI 0.05, 0.60), and those that always participated reported a 1.30-point higher rating (p<0.001, 95% CI 0.89, 1.70) (table 2). The always participated group also showed a 0.97-point higher rating compared with the occasionally participated group (p<0.001, 95% CI 0.55, 1.39). Higher ratings were found among hospitals with the following covariates: non-teaching versus teaching, small versus medium or large size, high versus low specialised service capability, Midwest versus other regions, metropolitan versus non-metropolitan areas, not-for-profit versus for-profit and hospitals without versus with system membership.

Compared with hospitals that never participated in the HEI, those that occasionally participated reported a 0.49-point higher patient recommendation score (p=0.011, 95% CI 0.11, 0.87), and those that always participated reported a 1.90-point higher score (p<0.001, 95% CI 1.36, 2.44). The always participated group also showed a 1.41-point higher recommendation score compared with the occasionally participated group (p<0.001, 95% CI 0.85, 1.96). Hospital covariates with higher recommendation scores were similar to those with higher overall ratings.

Performance in the HEI (hypothesis 2)

A 10-point increase in the total HEI score was associated with a 0.10-point increase in patient hospital ratings (p=0.031, 95% CI 0.01, 0.20) (table 3). Large hospitals had lower ratings than small hospitals. Hospitals with high specialised service capability scored higher than those with low capability. Northeast hospitals scored lower than those in the South, and for-profit hospitals had lower ratings than not-for-profit hospitals.

A 10-point increase in the total HEI score was associated with a 0.15-point increase in patient recommendations (p=0.023, 95% CI 0.02, 0.28). Hospital covariates with higher recommendations scores were similar to those with higher overall ratings.

Sensitivity analyses

Sensitivity analyses (online supplemental appendices D through G) revealed that LGBTQ+ inclusive patient

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services were associated with higher patient ratings (p=0.005, 95% CI 0.08, 0.46) and higher recommendations (p=0.004, 95% CI 0.12, 0.64). However, the subscores for non-discrimination policy and staff training in LGBTQ+ care, inclusive employee benefits and LGBTQ+ community outreach were not significantly associated with either outcome.

DISCUSSION

Our findings highlight the benefits of LGBTQ+ inclusion efforts on patient experience. Aligned with theories on inclusive organisations,[13 14] hospitals consistently participating in the HEI received greater patient satisfaction compared with those with no or intermittent participation, even when controlling for structural, operational and geographical covariates. While the HEI participation process is administrative in nature, it requires significant planning and implementation that may be linked to both organisational culture and patient care practices. Given the demonstrated patient benefits, hospital leadership should actively promote engagement in the HEI as a tool to guide hospitals towards a high degree of LGBTQ+ inclusive practices.

Consistent with our hypothesis, among hospitals participating in the HEI, incremental improvements in hospitals' total HEI scores were associated with small but significant increases in patient satisfaction. Implementing comprehensive LGBTQ+ inclusive practices may mediate improved satisfaction scores by fostering greater trust among LGBTQ+ patients, encouraging open communication and enhancing their perception of individualised care.[22] Inclusivity can also benefit the broader patient population by promoting a culture of respect and equity. Similar associations between LGBTQ+ inclusion and positive outcomes have been observed in non-healthcare settings, such as improved innovation and performance in corporations.[23 24] The effect size was smaller when HEI performance scores, which included only hospitals that participated in the HEI, were used as a predictor, compared with when HEI participation (which included all hospitals) was used as a predictor. This suggests that HEI participation is associated with how patients perceive their healthcare experience, regardless of performance scores. The smaller effect size may indicate that sustained HEI involvement is as important as performance fluctuations, as it reflects a broader institutional commitment to inclusivity.

We found that LGBTQ+ inclusive patient services were positively associated with patient satisfaction. These services, such as gender-affirming care, directly enhance the patient experience.[3] However, the arrival of a new US presidential administration signalling hostility towards LGBTQ+ populations, a federal-level anti-LGBTQ+ mandate[25] and a surge in state-level anti-LGBTQ+ laws,[26] particularly those restricting LGBTQ+ healthcare,[27] present barriers to care and threaten progress in inclusivity. In 2021 and 2023, 231 hospitals received HEI point deductions for discriminatory practices. As health organisations continue to stress health equity,[28] hospitals must advocate for policy changes and strengthen inclusivity efforts to support LGBTQ+ patients. Additionally, limited research has examined the organisational impact of LGBTQ+ inclusion on broader patient or clinical outcomes. Most studies focus on provider-level outcomes, such as LGBTQ+-specific training,[29 30] or exclusively on LGBTQ+ populations,[31 32] limiting generalisability. Future research should investigate the broader impact of organisational inclusion efforts on patient and clinical outcomes across diverse populations.

We also observed a decline in HCAHPS scores, particularly in 2021 during the COVID-19 pandemic, when staffing shortages, absenteeism and resource constraints likely affected patient satisfaction. Previous studies have shown similar drops, particularly in hospitals with fewer resources.[33] Although pandemic-related factors were not explicitly considered in our study, they may have contributed to variability in patient satisfaction. Future research should consider staffing levels and COVID-19 case rates to better assess their impact on hospital performance and patient experiences.

This study has several limitations. First, while it included a large US national sample, generalisability may be limited by national-level and state-level differences in healthcare delivery and political contexts. The exclusion of non-IPPS hospitals and those with low survey completion rates further limits generalisability. Second, we could not examine differential effects of LGBTQ+ inclusion between LGBTQ+ and non-LGBTQ+ patients due to the aggregated nature of HCAHPS data. Third, unmeasured confounders, such as hospitals prioritising diversity, equity and inclusion initiatives independent of HEI participation, may have influenced results. Fourth, the absolute differences in HCAHPS scores were small, with unclear clinical significance. However, hospitals are motivated to achieve higher HCAHPS scores to be eligible for greater financial reimbursements.[34] Additionally, even modest changes in patient satisfaction can accumulate over time, enhancing a hospital's public image and making it more appealing to potential patients. In competitive healthcare markets, where hospitals showcase their HCAHPS scores on their websites,[35 36] these incremental gains can strengthen a hospital's reputation and financial success. Lastly, although we used longitudinal data, the marginal model in our GEE analyses limits causal inferences. While GEE accounts for within-subject correlations in longitudinal data and generates population-averaged effects, it does not establish temporal causality. Future research should examine whether HCAHPS scores improve over time with consistent HEI participation.

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CONCLUSION

Hospitals reporting engagement in LGBTQ+ inclusion efforts are found to have higher patient satisfaction scores. As healthcare systems strive to address disparities and promote equity, integrating LGBTQ+ inclusive policies and practices may enhance quality care and improve patient experience outcomes.

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Contributors HY, MDM, JAB and HB made contributions to the conception and design of this article. HY, MDM, JAB and HB contributed to the acquisition, analysis and interpretation of data. HY drafted the manuscript. HY, MDM, JAB, TH and HB revised the manuscript. All authors approved the final manuscript. HY is the guarantor.

Funding The primary author (HY) received postdoctoral fellowship funding from the Eidos LGBTQ+ Health Initiative at the University of Pennsylvania.

Map disclaimer The inclusion of any map (including the depiction of any boundaries therein), or of any geographic or locational reference, does not imply the expression of any opinion whatsoever on the part of BMJ concerning the legal status of any country, territory, jurisdiction or area or of its authorities. Any such expression remains solely that of the relevant source and is not endorsed by BMJ. Maps are provided without any warranty of any kind, either express or implied.

Competing interests TH holds a leadership position with the Human Rights Campaign. The remaining authors declare no conflicts of interest.

Patient consent for publication Not applicable.

Ethics approval This study received an exemption from the Institutional Review Board.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data may be obtained from a third party and are not publicly available. Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) data are available on the publicly accessible Hospital Compare website. Healthcare Equality Index data were obtained from the Human Rights Campaign Foundation, and American Hospital Association (AHA) data were sourced directly from the AHA. Requests for these data should be directed to the respective organisations.

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